2019春高数下期中试题(回忆版)

1. Determine whether the following statements are true or false? No justification is necessary.

(1) If $a_n<0$ for $n>1000$, and $\sum_{n=1}^{\infty}a_n$ converges, then $\sum_{n=1}^{\infty}a_n^2$ may diverge.

(2) Suppose that the power series $\sum_{n=1}^{\infty}a_n(x-1)^n$ converges at $x=0$, and disverges at $x=2$, then the interval of convergence of this series is $[0,2)$.

(3) If $f(x,y)$ has partial derivatives $f_x(x,y)$ and $f_y(x,y)$ at (x_0,y_0), then
$$\lim_{x\to x_0}f(x,y_0)=\lim_{y\to y_0}f(x_0,y)=f(x_0,y_0).$$

(4) If a vector function $r(t)$ is always perpendicular to its derivative $\frac{dr}{dt}$, then $|r(t)|$ must be constant.

(5) The curvature of a unit circle is greater than the curvature of the parabola $y=x^2$ at the origin.

2. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) Let a and b be two nonzero orthogonal vectors, which of the following must be true?

(A) $|a+b|=|a|+|b|$; (B) $|a-b|=|a|-|b|$;

(C) $|a+b|=|a-b|$; (D) $a+b=a-b$.

(2) The equations of two lines are $l_1: x=t, y=2t, z=-t$, and $l_2: x=1-2t, y=t, z=-1+t$. Then l_1 and l_2 are

(A) parallel; (B) orthogonal; (C) intersect with each other; (D) skew.

(3) Suppose $0\le a_n\le\frac{1}{n}$, $(n=1,2,\cdots)$, then which of the following series must converge?

(A) $\sum_{n=1}^{\infty}a_n$; (B) $\sum_{n=1}^{\infty}(-1)^n a_n$; (C) $\sum_{n=1}^{\infty}\sqrt{a_n}$; (D) $\sum_{n=1}^{\infty}(-1)^n a_n^2$.

3. Does the following series absolutely converge, conditionally converge, or diverge? Give reasons for your answer.

(1) $\sum_{n=1}^{\infty}(-1)^n\left(1-\frac{1}{n}\right)^{n^2}$

(2) $\sum_{n=1}^{\infty}(-1)^n e^{-\frac{1}{n}}$.

(3) $\sum_{n=2019}^{\infty}(-1)^n\frac{1}{\sqrt{n^2-2019n+1}}$

4. (1) Find the radius and interval of convergence of the series
$$\sum_{n=0}^{\infty}\frac{(-1)^{n+1}2^n x^n}{\sqrt{n^2+n+1}}$$

(2) For what values of x does the series converge absolutely, or conditionally?

5. Let $x=\cos^3 t$, $y=\sin^3 t$, where $0\le t\le\frac{\pi}{2}$, be a parametrization of a curve.

(1) Find the length of the curve.

(2) Find the area of the surface generated by revolving the curve about the x-axis.

6. Find the equation of the plane through the points $(2,-1,-1)$ and $(1,0,-1)$ perpendicular to the plane $2x+3y-5z+6=0$.

7. A particle is located at the point $(1,0,-2)$. Its initial speed is $|v(0)|=3$ at time $t=0$, and the direction of its initial velocity is toward the point $(2,-1,3)$. The particle moves with constant acceleration $i+2j+k$. Find its position vector $r(t)$ at time t.

8. Is the following function, $f(x,y)$ continuous at $(0,0)$? Give reasons for your answer.

$$f(x,y)=\begin{cases}\dfrac{\sin(x^3+y^3)}{x^2+y^2}, & (x,y)\neq(0,0);\\ 0, & (x,y)=(0,0).\end{cases}$$

9. Let $f(u)$ be differentiable, $z=f(e^x y)$ $(y\neq 0)$, and $\dfrac{\partial z}{\partial x}+y\dfrac{\partial z}{\partial y}=1$. If $f(1)=0$, find $f(u)$.

10. Find the Taylor series for $f(x)=\ln(x+\sqrt{x^2+1})$ at $x=0$.